Theme Warm

Genomic Prediction

Advanced Statistics Lab

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Theme

Genomic Prediction

Genomic Prediction entails methods to predict breeding values based on phenotypic and molecular (SNP) data.

$$\hat{\mathbf{g}} = \mathbf{M}\hat{\mathbf{a}}$$

where $\mathbf{M}$ is a matrix of genotype scores in {0,1,2} coding and $\hat{\mathbf{a}}$ is a vector of average allele effects for those markers.

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Estimated Breeding Value

The estimated breeding value ($\hat{\mathbf{g}}$, EBV) is the sum of all the average allele effects for the marker alleles the individual carries.

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Ridge Regression BLUP

The average allele effects can be estimated directly in a mixed effects model, treating the them as random effects:

$$\mathbf{y} = \mu + \mathbf{Ma} + \mathbf{e}$$

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Center Marker Covariates

If the individuals in $\mathbf{M}$ are not representative of the general population, we have to adjust the breeding values obtained from the marker effects by the population mean.

$$\hat{\mathbf{g}} = \mathbf{M}\hat{\mathbf{a}} - 2\mathbf{p}\hat{\mathbf{a}}$$

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GBLUP

Instead of doing regression on the marker covariates directly, we can also generate a Gaussian Kernel from those covariates.

$$\mathbf{G} = \frac{\bar{\mathbf{M}}\bar{\mathbf{M}}'}{tr(\bar{\mathbf{M}}\bar{\mathbf{M}}')/n}$$

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Equivalent Models

The GBLUP model becomes:

$$\mathbf{y} = \mu + \mathbf{g} + \mathbf{e}$$

with $\mathbf{g} \sim MVN(0, \mathbf{G}\sigma_g^2)$.

Both models, the ridge regression marker model and GBLUP are statistically identical and result in the same EBVs.

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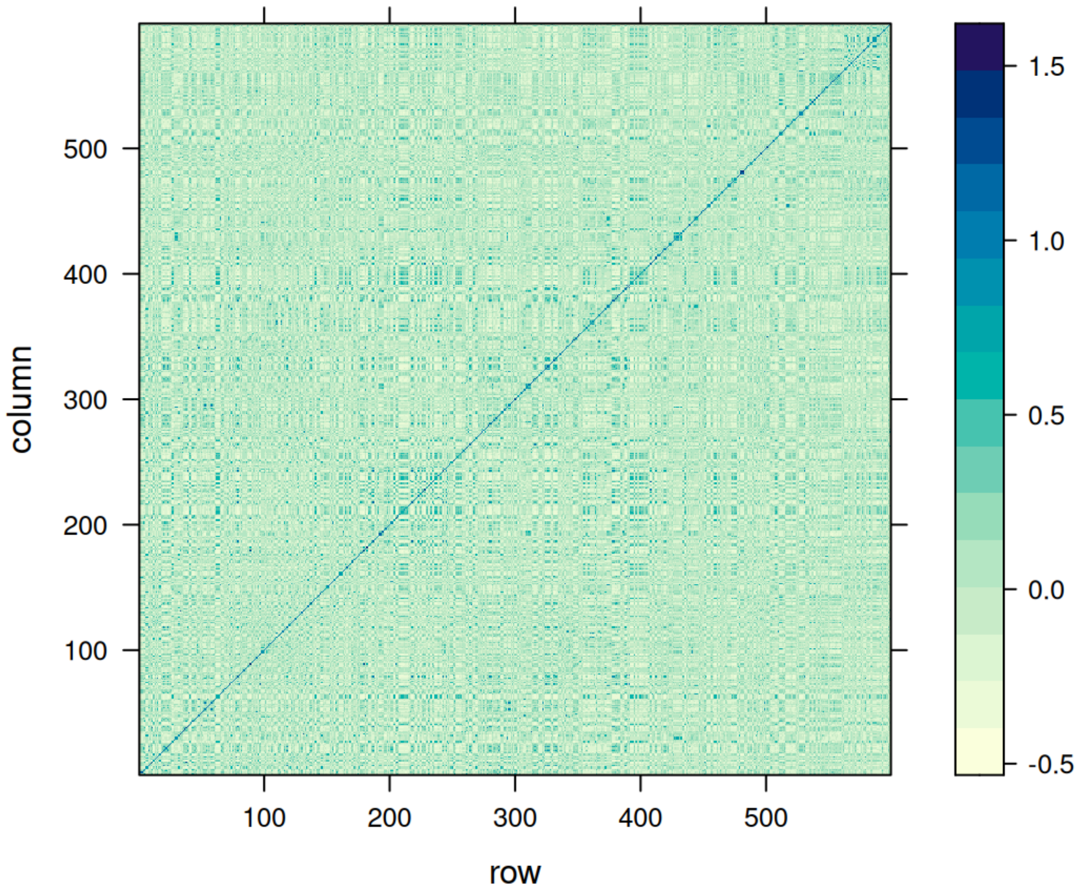
Prepare Wheat Data

```
library(BGLR)
library(rrBLUP)
library(ggplot2)
library(knitr)
library(lattice)
theme_set(theme_minimal(base_size = 16))

# load the "wheat" data set from BGLR
data(wheat)

# extract one phenotype
y = wheat.Y[,1]

# the marker covariates
M = wheat.X

# average allele frequencies ("p")
p = colMeans(M) / 2
```

Genomic Relationship Matrix



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Ridge Regression Marker Model

```
# design matrix for random effects (marker covariates become design matrix)
Z = Mc

# design matrix for fixed effects (only intercept)
X = array(1, dim = c(length(y), 1))

# fit the model with ML
mod_rr = mixed.solve(
         y = y,
         X = X,
         Z = Z,
         method = "ML"
         )
```

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Marker Effects And EBV

```r
# extract marker effects
a_rr = mod_rr$u

# predict breeding values from the model
g_rr = as.numeric(Z %*% a_rr)

# extract variance components
var_a_rr = mod_rr$Vu
var_e_rr = mod_rr$Ve

# rescale the marker-effect variance component
var_g_rr = mod_rr$Vu * mean(diag(MMdash))
```

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GBLUP Model

```r
# fit the model with ML
mod_gblup = mixed.solve(
          y = y,
          K = G,
          method = "ML"
          )

# extract breeding values
g_gblup = mod_gblup$u
```

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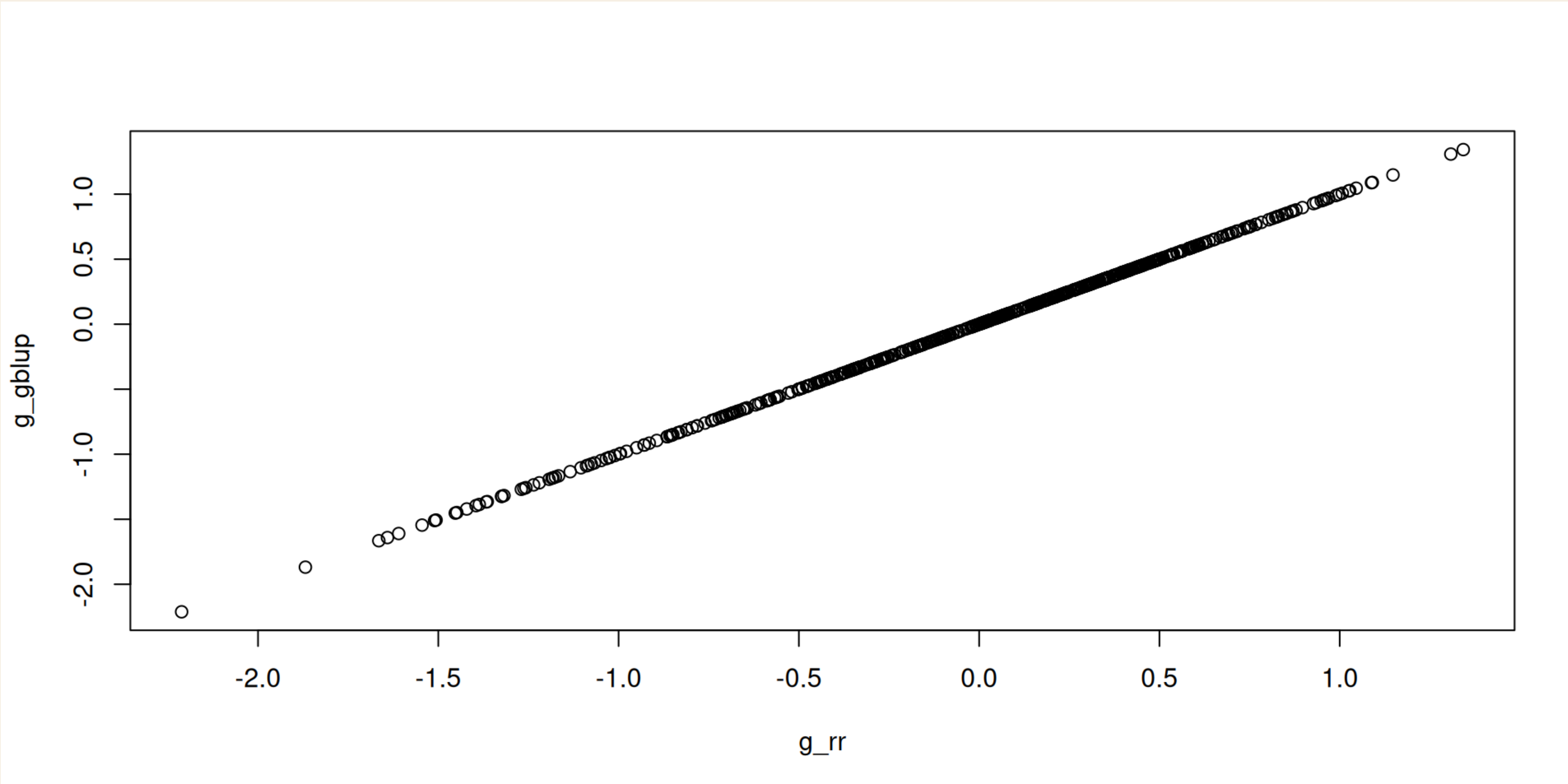
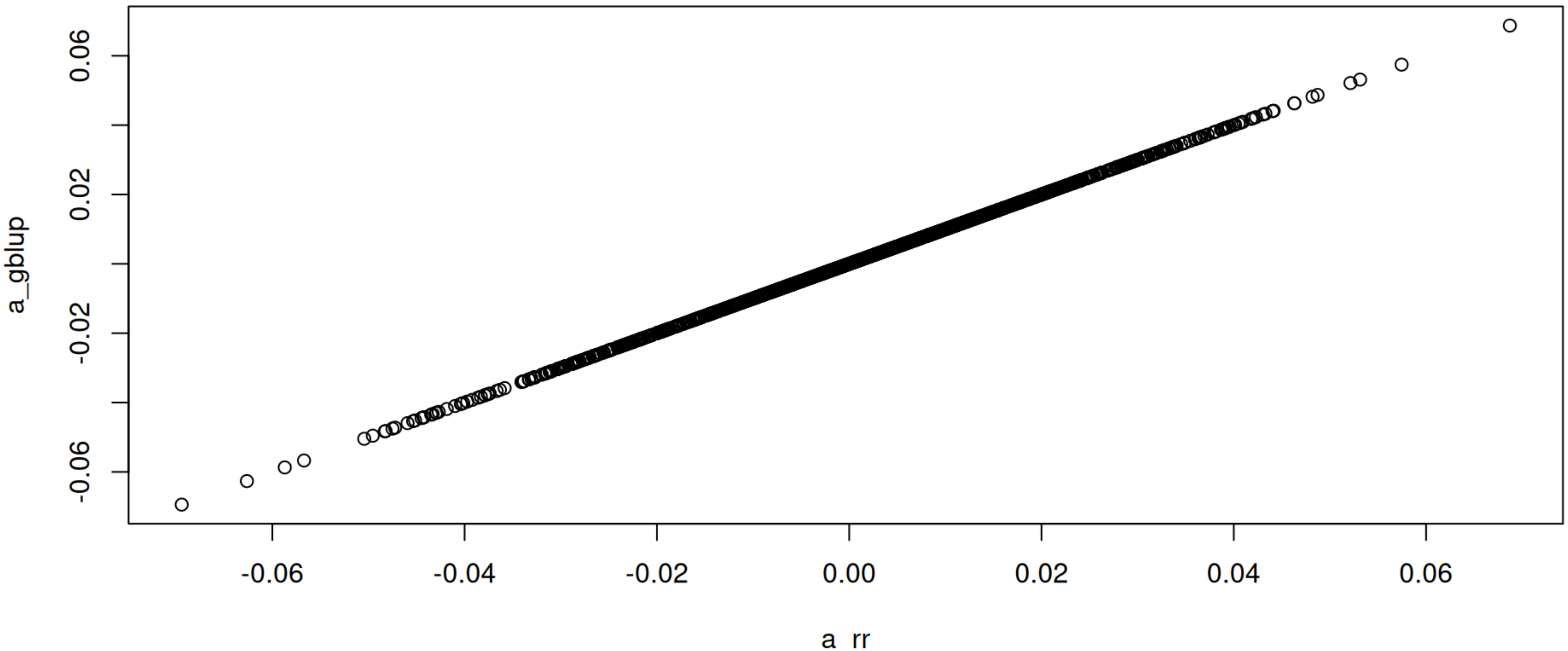
Marker Effects From GBLUP

```
MMdash_tmp = MMdash
diag(MMdash_tmp) = diag(MMdash_tmp) + 0.00001
a_gblup = as.numeric(t(Mc) %*% solve(MMdash_tmp, g_gblup))

# extract variance components
var_g_gblup = mod_gblup$Vu
var_e_gblup = mod_gblup$Ve
```

Check Equivalence




a_gblup
0.06
0.02
-0.02
-0.06
-0.06
-0.04
-0.02
0.00
0.02
0.04
0.06
a_rr

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Variance Components

```r
vc = rbind(
        data.frame(
         model = "Marker Model",
         var_g = var_g_rr,
         var_e = var_e_rr,
         h2 = var_g_rr / (var_g_rr + var_e_rr)
         ),
        data.frame(
         model = "GBLUP",
         var_g = var_g_gblup,
         var_e = var_e_gblup,
         h2 = var_g_gblup / (var_g_gblup + var_e_gblup)
         )
        )

kable(vc, digits = 3)
```

model	var_g	var_e	h2
Marker Model	0.605	0.539	0.529
GBLUP	0.605	0.539	0.529

Cross Validation

In order to assess the performance of our genomic prediction models, we can run a cross validation scheme, in which we train marker effects with a subset of the data and predict the EBVs of the left out samples based on the training model.

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Cross-Validation Helper

```r
cCV <- function(y, folds=5, reps=1, matrix=FALSE, seed=NULL) {

  if(!is.vector(y)) stop("y must be a vector")
  if(folds < 2) stop("'folds' must be larger than one")
  if(reps < 1) reps = 1

  reps = as.integer(reps)
  folds = as.integer(folds)

  isy <- (1:length(y))[!is.na(y)]

  n_vec <- rep(NA,folds)
  start <- rep(NA,folds)
  end <- rep(NA,folds)

  n_vec[1:(folds-1)] <- round(length(isy)/folds,digits=0)
  n_vec[folds] <- length(isy) - sum(n_vec[1:(folds-1)])

  for(j in 1:folds) {

    if(j==1) {
      start[j] = 1
    } else {
      start[j] = end[j-1]+1
    }
```

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Cross-Validation Fit

```
cvs = cCV(y, folds = 5, reps = 10)

preds = list()
for(i in 1:length(cvs)) {

  this_pred = mixed.solve(y = cvs[[i]], K = G)$u[is.na(cvs[[i]])]
  this_y = y[is.na(cvs[[i]])]
  preds[[i]] = data.frame(pred = this_pred, y = this_y)

}

cors = unlist(lapply(preds, function(x) cor(x$pred, x$y)))
```

Bayesian Alphabet

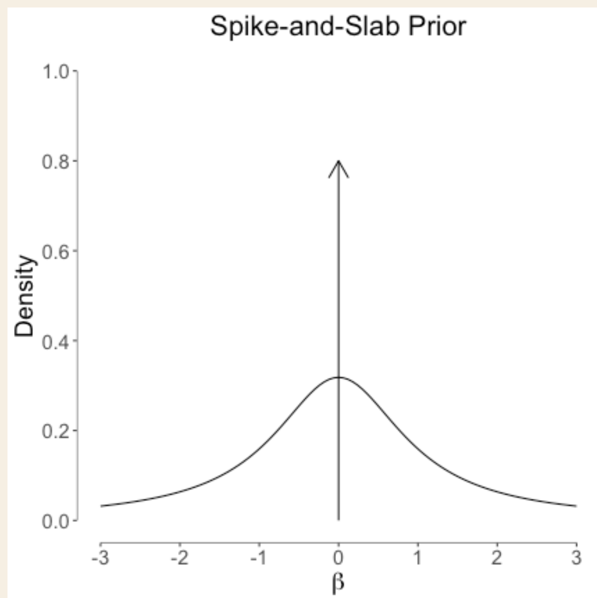
A linear mixed model with a series of independent random effects has the following form:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}_1\mathbf{u}_1 + \mathbf{Z}_2\mathbf{u}_2 + \mathbf{Z}_3\mathbf{u}_3 + \cdots + \mathbf{Z}_k\mathbf{u}_k + \mathbf{e}$$

Spike-And-Slap

In the Bayesian Alphabet, the prior distributions on the marker effects is altered to generally generate a sparsity pattern on the marker effects, meaning feature selection.

This feature selection is achieved by using *spike-and-slap* mixture priors.



Lab notes

Theme

Priors

Table 1. Summary of some effect size distributions that have been proposed for polygenic modeling.

Effect Size Distribution		Keywords	Selected References
Name	**Formula**		
t	$\beta_i \sim t(0,v,\sigma_a^2)$	BayesA	[27,32,33,40]
point-t	$\beta_i \sim \pi t(0,v,\sigma_a^2)+(1-\pi)\delta_0$	BayesB, BayesD, BayesDπ	[27,32–34,40]
t mixture	$\beta_i \sim \pi t(0,v,\sigma_a^2)+(1-\pi)t(0,v,0.01\sigma_a^2)$	BayesC	[32,33]
point-normal	$\beta_i \sim \pi \mathrm{N}(0,\sigma_a^2)+(1-\pi)\delta_0$	BayesC, BayesCπ, BVSR	[18,19,34]
double exponential	$\beta_i \sim \mathrm{DE}(0,\theta)$	Bayesian Lasso	[28,39,68]
point-normal mixture	$\beta_i \sim \pi_1 \mathrm{N}(0,\sigma_a^2)+\pi_2 \mathrm{N}(0,0.1\sigma_a^2)+\pi_3$ $\mathrm{N}(0,0.01\sigma_a^2)+(1-\pi_1-\pi_2-\pi_3)\delta_0$	BayesR	[35]
normal	$\beta_i \sim \mathrm{N}(0,\sigma_a^2)$	LMM, BLUP, Ridge Regression	[22,26,28,48]
normal-exponential-gamma	$\beta_i \sim \mathrm{NEG}(0,\kappa,\theta)$	NEG	[16]
normal mixture	$\beta_i \sim \pi \mathrm{N}(0,\sigma_a^2+\sigma_b^2)+(1-\pi)\mathrm{N}(0,\sigma_b^2)$	BSLMM	Present Work

The reference list contains only a selection of relevant publications. Abbreviations: DE denotes double exponential distribution, NEG denotes normal exponential gamma distribution, and other abbreviations can be found in the main text. In the scaled t-distribution, v and σ_a^2 are the degree of freedom parameter and scale parameter, respectively. In the DE distribution, θ is the scale parameter. In the NEG distribution, κ and θ are the shape and scale parameters, respectively. Notes: 1. Some applications of these methods combine a particular effect size distribution with a random effects term, with covariance matrix K, to capture sample structure or relatedness. If $K \propto XX^T$ then this is equivalent to adding a normal distribution to the effect size distribution. The listed effect size distributions in this table do not include this additional normal component. 2. BayesC has been used to refer to models with different effect size distributions in different papers. 3. In some papers, keywords listed here have been used to refer to fitting techniques rather than effect size distributions.

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Theme

$BayesC\pi$ - Threshold Model

$$
\begin{aligned}
\mathbf{y} &\sim \mathbf{B}(1, \mathbf{p}) \\
\mathbf{p} &= probit(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}) \\
\boldsymbol{\beta} &\sim \mathbf{U}(-Inf, Inf) \\
\mathbf{u} &\sim \pi N(0, {\sigma_u}^2) + (1 - \pi)\delta_0 \\
\pi &\sim U(0, 1) \\
{\sigma_u}^2 &\sim Gamma(1, 0.01)
\end{aligned}
$$